MOSAIC: Data-Certified Stochastic Release under Structured Deployment Shift Supplementary Material

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1 Notation and Exact Support Functions

For one task label, let $P = (p_1, \ldots, p_G)$ collect the conditional fine-token laws and let $M \in \mathrm{Stoch}(K, L)$ be a source-blind release channel. For an attacker assignment $a \in [G]^L$, define

$$v_{M,a,s}(c) = \sum_{z=1}^{L} M(c,z)\mathbf{1}\{a(z) = s\}. \tag{1}$$

Every $v_{M,a,s}$ lies in $[0,1]^K$. The balanced Bayes accuracy has the two equivalent forms

$$\begin{aligned} A(P,M) &= \frac{1}{G}\sum_{z}\max_{s}(p_s M)(z) \\ &= \frac{1}{G}\max_{a\in[G]^L}\sum_{s} p_s^\top v_{M,a,s}. \end{aligned} \tag{2}$$

The second identity follows because a finite-alphabet Bayes rule can be chosen deterministically at each released token.

For empirical law $\widehat{p}$, radius ϵ, and bounded score v, the confidence-set support function is

$$\Phi(\widehat{p}, \epsilon; v) = \max_{p\geq 0,\, \mathbf{1}^\top p=1,\, \|p-\widehat{p}\|_1\leq\epsilon} p^\top v. \tag{3}$$

Introducing positive and negative displacement variables gives a linear program. Its compact dual is

$$\min_{\nu,\lambda,\theta} \quad \nu + \widehat{p}^\top\theta + \epsilon\lambda, \tag{4}$$

$$\text{s.t.} \quad \nu + \theta_c \geq v_c \quad \forall c, \tag{5}$$

$$-\lambda \leq \theta_c \leq \lambda \quad \forall c, \qquad \lambda \geq 0. \tag{6}$$

Strong duality holds because the primal confidence set is nonempty and compact. The implementation also evaluates Eq. (3) independently by moving at most $\epsilon/2$ probability mass from low-score to high-score coordinates, respecting simplex boundaries.

2 Adaptive Confidence Envelope

Let $\mathcal{C}_s = \{p \in \Delta_K : \|p - \widehat{p}_s\|_1 \leq \epsilon_s\}$ and define

$$\overline{A}(\widehat{P}, M) = \frac{1}{G}\max_{a\in[G]^L}\sum_{s}\Phi(\widehat{p}_s, \epsilon_s; v_{M,a,s}). \tag{7}$$

Theorem 1 (Exact adaptive universal-attacker envelope)**.** *On the event $\mathcal{E} = \{p_s \in \mathcal{C}_s\ \forall s, y\}$, $A(P,M) \leq \overline{A}(\widehat{P}, M)$ simultaneously for every stochastic release channel M, including one selected from the same table. For fixed M, $\overline{A}$ is the exact supremum of $A(P,M)$ over $\mathcal{C}_1 \times \cdots \times \mathcal{C}_G$.*

Proof. Using Eq. (2), take the supremum over the product confidence set. The assignment family is finite, so the supremum and assignment maximum commute. For a fixed assignment, the objective is additive in sources and the confidence set is a Cartesian product, so each source maximization separates and equals the support function in Eq. (3). This proves exactness for every fixed M. On $\mathcal{E}$, the true law is one feasible point in every confidence set, giving the upper bound. The statement is pointwise over the complete channel space, so it remains true after any measurable same-table selection of M. □

Corollary 2 (Unconditional and label-aware attackers)**.** *Let $X = (Y, C)$ and let r_s denote the natural, unbalanced law $P(X \mid S = s)$. Define the lifted channel $\widetilde{M}((y,c),z) = M(c,z)$. Applying the theorem to $(r_1, \ldots, r_G)$ certifies every source attacker observing Z without conditioning on Y. Replacing the output by (Y, Z) and using $\widetilde{M}((y,c),(y',z)) = \mathbf{1}\{y = y'\}M(c,z)$ certifies the stronger attacker that also observes the true label.*

The corollary requires confidence regions for the natural joint laws and is not implied by source–label-balanced sampling. The primary experiments instead register the maximum label-conditional advantage, which removes the prevalence channel by design.

For conditionally i.i.d. categorical observations within each stratum, positive fixed count $n_{s,y}$, and an outcome-independent allocation $\delta_{s,y}$, the confirmation uses

$$\epsilon_{s,y} = \min\left\{2, \sqrt{\frac{2\log((2^K-2)/\delta_{s,y})}{n_{s,y}}}\right\}. \tag{8}$$

A union bound gives $\Pr(\mathcal{E}) \geq 1 - \sum_{s,y} \delta_{s,y}$. A zero-count stratum receives radius two and is covered deterministically. The locked bridge protocol used "exchangeable" informally for randomized held-out rows from the same stratum; arbitrary exchangeability alone does not imply this concentration bound, and is not claimed here.

3 Data-Certified Shift Bridge

For each source and task label, let $\mathcal{C}_{s,y}$ be the reference confidence set above and let

$$\mathcal{D}_{s,y} = \{q \in \Delta_K : \|q - \widehat{q}_{s,y}\|_1 \leq \xi_{s,y}\} \tag{9}$$

be the confidence set from a labeled target bridge fold. Define

$$h_{s,y}(w) = \max_{p \in \mathcal{C}_{s,y}} p^\top w, \quad \underline{q}_{s,y}(z) = \min_{q \in \mathcal{D}_{s,y}} q(z). \tag{10}$$

The coordinate lower bound is $-\Phi(\widehat{q}_{s,y}, \xi_{s,y}; -e_z)$ and is therefore exact. For fixed y, solve

$$\begin{aligned} \widehat{t}_y = \max_{t,W} \quad & t \\ \text{s.t.} \quad & W_{cz} \geq 0, \quad \sum_z W_{cz} = t \quad \forall c, \\ & h_{s,y}(W_{:z}) \leq \underline{q}_{s,y}(z) \qquad \forall s, z, \\ & 0 \leq t \leq 1. \end{aligned} \tag{11}$$

This is an LP. In particular, each support constraint is represented by variables $(\nu_{s,z}, \lambda_{s,z}, \theta_{s,z})$ satisfying

$$\begin{aligned} \nu_{s,z} + \theta_{s,z,c} &\geq W_{cz} && \forall c, \\ -\lambda_{s,z} \leq \theta_{s,z,c} &\leq \lambda_{s,z} && \forall c, \\ \nu_{s,z} + \widehat{p}_{s,y}^\top \theta_{s,z} + \epsilon_{s,y}\lambda_{s,z} &\leq \underline{q}_{s,y}(z), && \lambda_{s,z} \geq 0. \end{aligned} \tag{12}$$

Strong duality for Eq. (3) makes this epigraph exact.

Theorem 3 (Adaptive finite-sample bridge membership)**.** *Suppose the reference and bridge sets jointly cover all $p_{s,y}$ and $q_{s,y}$. Any feasible (t, W) in Eq.* (11)*, including its same-table maximizer, satisfies, for $t > 0$,*

$$q_{s,y} = t p_{s,y} T + (1-t) r_{s,y}, \qquad T = W/t, \tag{13}$$

for one source-blind stochastic T and source-specific probability vectors $r_{s,y}$.

Proof. Coverage and feasibility give, for every source and output coordinate,

$$q_{s,y}(z) \geq \underline{q}_{s,y}(z) \geq h_{s,y}(W_{:z}) \geq p_{s,y}^\top W_{:z}. \tag{14}$$

Thus $d_{s,y} = q_{s,y} - p_{s,y} W$ is coordinatewise nonnegative. Since every row of W sums to t, $\mathbf{1}^\top d_{s,y} = 1 - t$. If $t < 1$, set $r_{s,y} = d_{s,y}/(1-t)$; if $t = 1$, then $d_{s,y} = 0$ and any residual law may be used. Also, $T = W/t$ is row-stochastic. Every implication above holds simultaneously for every feasible (t, W) on one event, so choosing the maximizer from the same confidence tables costs no additional multiplicity. ☐

Proposition 4 (Robust optimality of the bridge LP)**.** *Over the Cartesian product of all reference and bridge confidence sets, Eq.* (11) *returns the largest retained mass for which one fixed source-blind transform certifies Eq.* (13) *uniformly over every law in that product.*

Proof. For fixed (t, T), the decomposition exists for a particular (p, q) exactly when $q_{s,y} - t p_{s,y} T$ is coordinatewise nonnegative: its coordinates then sum to $1 - t$ and can be normalized into a residual law. Requiring this for all (p, q) in the product region is equivalent to

$$\max_{p \in \mathcal{C}_{s,y}} t(pT)(z) \leq \min_{q \in \mathcal{D}_{s,y}} q(z) \quad \forall s, z. \tag{15}$$

The two extrema separate because the region is a Cartesian product. Substituting $W = tT$ gives exactly the constraints of Eq. (11); Eq. (12) introduces no relaxation. Maximizing t proves the claim. ☐

Corollary 5 (End-to-end target-population certificate)**.** *Suppose $\widehat{t}_y > 0$ for every required label. Insert the singleton transform $\{\widehat{T}_y\}$ and contamination $\widehat{\eta}_y = 1 - \widehat{t}_y$ from the bridge LP into the transform-exact within-label source-distinguishability and utility optimizer. On the same joint event, every same-table-selected release channel and decoder obeys its certified contracts on the bridge population. This remains simultaneous across task labels, registered utility thresholds, and finitely many released alphabet sizes when their underlying reference and bridge tables share the registered allocation.*

Proof. The bridge theorem establishes membership in the structured class used by the transform-exact source-distinguishability and utility theorems below. Those theorems are pointwise over every release channel and decoder on the same reference event. Intersecting with the bridge confidence event and substituting the selected singleton transform proves the result. Thresholds and alphabet sizes index decisions made inside that already covered collection; separately learned tokenizers or erasers still require familywise allocation across their distinct tables. ☐

If a required bridge stratum has count zero, its honest set $\mathcal{D}_{s,y}$ is the full simplex and every $\underline{q}_{s,y}(z) = 0$. For any $t > 0$, the probability vector $p_{s,y} T$ has a positive coordinate, contradicting the corresponding bridge constraint. Hence $\widehat{t}_y = 0$. This is the operational missing-source boundary used in the bridge confirmation.

4 Channel Capacity and Common-Shift Transfer

Define

$$\mathrm{Cap}_G(M) = \frac{1}{G} \max_{a \in [G]^L} \sum_s \max_c v_{M,a,s}(c). \tag{16}$$

Lemma 6 (Exact arbitrary-shift capacity)**.** $\mathrm{Cap}_G(M) = \sup_{r_1, \ldots, r_G \in \Delta_K} A(R, M)$.

Proof. Expand $A(R, M)$ by Eq. (2). For fixed assignment a, each source law r_s can concentrate all mass on a token maximizing $v_{M,a,s}(c)$. The source suprema separate and the finite maxima commute, yielding Eq. (16). ☐

For two sources, uniform-prior Bayes accuracy is $(1 + \mathrm{TV}(r_1M, r_2M))/2$. The diameter of the convex hull of channel rows is attained by two rows, hence $2\,\mathrm{Cap}_2(M) - 1 = max_{c,c'}\,\mathrm{TV}(M(c,\cdot), M(c',\cdot))$, the Dobrushin coefficient.

For common pre-release channel T, define

$$\rho_T(M) = \max_c \mathrm{TV}((TM)(c,\cdot), M(c,\cdot)). \tag{17}$$

Lemma 7 (Approximate invariance transfer)**.** *For every source experiment P,*

$$A(P, TM) \le \min\{\mathrm{Cap}_G(M), A(P, M) + \rho_T(M)\}. \tag{18}$$

Proof. For source s, convexity of total variation gives $\mathrm{TV}(p_sTM, p_sM) \le \sum_c p_s(c)\,\mathrm{TV}((TM)(c,\cdot), M(c,\cdot)) \le \rho_T(M)$. The success probability of any attacker changes by at most this distance. Averaging over sources and maximizing the attacker gives the additive bound. The capacity bound follows by treating p_sT as an arbitrary fine-token law and applying the preceding lemma. □

5 Structured External Shift

For label y, suppose

$$q_s = t_y p_s T_y + (1 - t_y) r_s, \qquad t_y \ge 1 - \eta_y, \tag{19}$$

with common $T_y \in \mathcal{T}_y$ and common retained mass t_y across sources. For a polytope with finite extremes $\mathcal{V}_y = \mathrm{ext}(\mathcal{T}_y)$, define

$$\begin{aligned}\overline{A}_y^X(M) = \frac{1}{G} \max_{T\in\mathcal{V}_y,\, a\in[G]^L} \sum_s \Big[\\ (1-\eta_y)\Phi(\widehat{p}_{s,y}, \epsilon_{s,y}; Tv_{M,a,s}) \\ + \eta_y \max_c v_{M,a,s}(c)\Big].\end{aligned} \tag{20}$$

Theorem 8 (Transform-exact post-selection certificate)**.** *On $\mathcal{E}$, simultaneously for every selected M and every external law in Eq.* (19)*, $A(Q, M) \le \overline{A}_y^X(M)$. For fixed M, the right-hand side is the exact supremum over the product confidence sets, transform polytope, allowed retained masses, and residual laws.*

Proof. Expand balanced Bayes accuracy using Eq. (2). For fixed T, t, and attacker assignment, the source confidence sets and residual simplexes separate. Their support values are $\Phi(\widehat{p}_{s,y}, \epsilon_{s,y}; Tv_{M,a,s})$ and $\max_c v_{M,a,s}(c)$. The latter is no smaller because every coordinate of Tv is a convex combination of coordinates of v. The maximizing retained mass is therefore $t = 1 - \eta_y$. The resulting confidence support is convex in T, so its maximum over a polytope is attained at an extreme. All remaining maxima are finite and commute. This gives exactness for fixed M; pointwise coverage over all channels preserves the bound after same-table selection. □

For a transform family that cannot be enumerated, let $B_y(M) = \min\{\mathrm{Cap}_G(M), \overline{A}_y(\widehat{P}, M) + \rho_{\mathcal{T}_y}(M)\}$.

Corollary 9 (Capacity-transfer fallback)**.** *On $\mathcal{E}$,*

$$A(Q, M) \le (1 - \eta_y) B_y(M) + \eta_y\,\mathrm{Cap}_G(M). \tag{21}$$

Proof. The retained experiment is bounded by Eq. (18), the adaptive envelope bounds its unknown reference term, and residuals are bounded by capacity. Since $B_y(M) \le \mathrm{Cap}_G(M)$, the largest value over $t_y \ge 1 - \eta_y$ occurs at $t_y = 1 - \eta_y$. □

The exact confidence envelope is no larger than this fallback because it is the supremum over the same set. At zero confidence radius, it becomes the known-population external risk

$$\begin{aligned}\frac{1}{G} \max_{T\in\mathrm{ext}(\mathcal{T}_y),\, a\in[G]^L} \sum_s \big[(1-\eta_y) p_{s,y}^\top T v_{M,a,s} \\ + \eta_y \max_c v_{M,a,s}(c)\big].\end{aligned} \tag{22}$$

used by the separately implemented synthetic evaluator.

6 Utility Certificate

Fix decoder $g : [L] \to [J]$, let $\ell_y(z) = \mathbf{1}\{g(z) \ne y\}$, and write $u_y = M\ell_y$. Define

$$\overline{E}_{s,y}(M, g) = \Phi(\widehat{p}_{s,y}, \epsilon_{s,y}; u_y), \tag{23}$$

$$\kappa_y(M, g) = \sup_{T\in\mathcal{T}_y} \max\{0, \max_c((TM - M)\ell_y)(c)\}, \tag{24}$$

$$U_y(M, g) = \max_c u_y(c). \tag{25}$$

Define the transform-exact value

$$\begin{aligned}\overline{E}_{s,y}^X(M, g) = \max_{T\in\mathcal{V}_y} \Big[(1-\eta_y) \\ \Phi(\widehat{p}_{s,y}, \epsilon_{s,y}; Tu_y) + \eta_y \max_c u_y(c)\Big].\end{aligned} \tag{26}$$

Corollary 10 (Transform-exact adaptive utility certificate)**.** *On $\mathcal{E}$, every selected (M, g) satisfies $\mathrm{Err}_{s,y}(Q, M, g) \le \overline{E}_{s,y}^X(M, g)$, and the right-hand side is exact over the stated confidence ball and shift class.*

Proof. Repeat the transform-exact proof for bounded loss ℓ_y, without averaging over sources or maximizing an attacker assignment. □

The capacity fallback is

Corollary 11 (Adaptive shifted-utility fallback)**.** *On $\mathcal{E}$, every selected (M, g) satisfies*

$$\mathrm{Err}_{s,y}(Q, M, g) \le (1 - \eta_y) D_{s,y}(M, g) + \eta_y U_y(M, g), \tag{27}$$

where $D_{s,y} = \min\{U_y, \overline{E}_{s,y} + \kappa_y\}$.

Proof. Apply the structured-shift proof to the bounded loss ℓ_y. The common component differs from its reference expectation by at most the largest positive directional defect κ_y, while the arbitrary residual concentrates on a row with maximal error U_y. □

The transform-exact utility value is never larger than this fallback.

The exact population utility risk used in the independent evaluator is

$$\max_{T\in\mathrm{ext}(\mathcal{T}_y)} \Big[(1-\eta_y) p_{s,y}^\top T M \ell_y + \eta_y \max_c (M\ell_y)(c)\Big]. \tag{28}$$

7 No-Free-Lunch Boundary

Theorem 12 (Unrestricted differential shift)**.** *For $G \geq 2$, $\mathrm{Cap}_G(M) = 1/G$ if and only if all rows of M are identical.*

Proof. Identical rows make every released source law identical. Conversely, if two rows differ, choose two residual source laws concentrated on those rows and place every remaining source on either row. The released source laws are not all identical, so uniform-prior Bayes accuracy is strictly above chance. Exactness of Eq. (16) completes the argument. □

For two fine tokens c_0, c_1 with different task labels, suppose a decoder has row-wise errors at most e_0, e_1. The decoder decision event occurs with probability at least $1 - e_0$ under $M(c_0, \cdot)$ and at most e_1 under $M(c_1, \cdot)$. Total variation is therefore at least $1 - e_0 - e_1$, giving

$$2\,\mathrm{Cap}_2(M) - 1 \geq 1 - e_0 - e_1. \tag{29}$$

Thus a nontrivial task contract is incompatible with chance-level source capacity under unrestricted differential shift.

8 Repeated Queries and Side Information

Proposition 13 (Persistent-query equivalence)**.** *Suppose the runtime samples one token Z per immutable item and every later query returns that stored token. For any finite or countably infinite number of queries, the complete transcript has exactly the same source-inference Bayes accuracy as one observation of Z.*

Proof. Every persistent transcript has the form $(Z, Z, \ldots)$. The transcript is a deterministic function of Z, while Z is recovered from its first coordinate. They therefore generate the same sigma-field. The classes of measurable source decisions based on either observation coincide, so their optimal balanced accuracies are equal. □

Fresh independent draws are a different mechanism. For a registered budget r, define the product channel

$$M^{\otimes r}(c, z_{1:r}) = \prod_{j=1}^{r} M(c, z_j). \tag{30}$$

Proposition 14 (Exact bounded-query reduction)**.** *An attacker observing r conditionally independent fresh releases sees exactly one output of $M^{\otimes r}$. Substituting this product channel into the adaptive, bridge, structured-shift, and capacity results gives an exact certificate for that query budget, without a union bound over queries.*

Proof. Conditioned on fine token c, the transcript probability mass function is the right-hand side of Eq. (30). Every theorem depends on the release mechanism only through this conditional channel. Replacing the output alphabet $[L]$ by $[L]^r$ therefore reproduces the transcript experiment exactly. The alphabet has size L^r, which is a computational rather than a statistical restriction. □

Proposition 15 (Unbounded fresh-query failure)**.** *If two rows $M(c, \cdot)$ and $M(c', \cdot)$ differ, then their r-fold product laws have total variation tending to one. Consequently the binary source-inference capacity of $M^{\otimes r}$ tends to one as $r \to \infty$.*

Proof. For distinct finite-alphabet laws u and v, their Hellinger affinity $\alpha = \sum_z \sqrt{u(z)v(z)}$ is strictly below one. Product affinity equals α^r, and $\mathrm{TV}(u^{\otimes r}, v^{\otimes r}) \geq 1 - \alpha^r \to 1$. Concentrating two source laws on fine tokens c and c' makes binary balanced Bayes accuracy $(1 + \mathrm{TV})/2 \to 1$. The capacity is at least this value and at most one. □

The registered attacker observes the persistent token, knows the complete mechanism, and has no additional item-level side information. If a variable H is also public, the relevant experiment is (Z, H), not Z; it must be included in the fine/output alphabet and certified jointly. In particular, original features, source-dependent metadata, mutable identifier behavior, timing, and private mechanism state are outside the theorem. A production runtime must durably bind each versioned item identifier to one immutable fine token and one sampled output; rebinding or silent rerandomization violates the registered mechanism.

Persistence is an item-level statement, not a population-level composition theorem. Tokens from distinct items associated with one source form a joint experiment. Even when each item is persistent, independent observations from different items can increase source distinguishability, while correlations, adaptive mechanisms, stopping times, and observable query counts can change the joint law further. Such access requires a certificate for the complete multi-item transcript. The product-channel proposition applies only to repeated conditionally independent draws from one fixed fine token under one fixed channel and a noninformative registered query count.

9 Missing-Source Non-Identifiability

Fix two required sources and a released-token law $q_0 M$ for source zero. Suppose an external audit contains no source-one rows, so its observed-data distribution is identical for every $q_1 \in \Delta_K$. Define

$$A_{\mathrm{miss}}(q_0, M) = \sup_{q_1 \in \Delta_K} A((q_0, q_1), M). \tag{31}$$

Theorem 16 (Missing-source non-identifiability)**.** *Fix q_0, M, and τ independently of the missing-source law and the audit randomness. Let a possibly randomized protocol observe no source-one rows and claim $A((q_0, q_1), M) \leq \tau$. If its false-certification probability is at most $\delta < 1$ uniformly over every q_1, then for $\tau < A_{\mathrm{miss}}(q_0, M)$ it can issue that claim with probability at most δ.*

Proof. The finite-simplex supremum is attained; call a maximizer q_1^*. The audit's observed-data distribution does not depend on the missing law. Consequently, every event on which the protocol claims a bound below $A((q_0, q_1^*), M)$ is a false-certification event under q_1^*, and uniform validity limits that event to probability δ. □

The same indistinguishability argument for a missing source–label utility stratum replaces Eq. (31) by $\max_c (M\ell_y)(c)$. These are precisely the full-simplex terms used when a stratum has count zero. Moreover, if M is non-constant, then $A_{\mathrm{miss}}(q_0, M) > 1/2$ for every q_0: the mixture $q_0 M$ cannot equal every nonidentical channel row, and a missing source concentrated on a differing row has positive total-variation distinguishability. Missing required strata must therefore be reported as unestimable unless an external structural assumption identifies their laws.

10 Abstention Theory

Let $V(P, e)$ be the globally optimal certified worst conditional error for confidence balls centered at P with common radius e. With deployed radius e_n, define $r_n^* = \sup\{r \geq 0 : V(P, e_n + r) \leq \tau_U\}$.

Proposition 17 (Finite-sample abstention envelope)**.** *For H registered strata of size n,*

$$\Pr(\textit{MOSAIC abstains}) \leq \min\{1, H(2^K - 2)e^{-n(r_n^*)^2/2}\}. \tag{32}$$

Proof. If every empirical stratum is within distance r of its population law, each radius-e_n ball around the empirical law is contained in the radius-$(e_n + r)$ ball around the population law. A feasible population-centered pair for the larger balls remains feasible for the empirical problem. Apply the Weissman bound and a stratum union bound, then let r approach r_n^*. □

For the preregistered local power curve, write $V_n(P) = V(P, e_n)$. Under local uniqueness of the decoder and certificate branches together with strong regularity of the resulting parametric linear program, the value is differentiable. If $h_{n,s,y}$ is its simplex gradient and $\Sigma_{s,y} = \mathrm{diag}(p_{s,y}) - p_{s,y}p_{s,y}^\top$, the finite-$n$ delta-method prediction is

$$\Phi_{\mathrm{N}}\left(\frac{\tau_U - V_n(P)}{\sqrt{n^{-1}\sum_{s,y} h_{n,s,y}^\top \Sigma_{s,y} h_{n,s,y}}}\right). \tag{33}$$

The locked analysis estimates each gradient by centered simplex-preserving finite differences at step sizes 10^{-4} and 5×10^{-5} and rejects the approximation if their maximum discrepancy exceeds 5×10^{-3}.

11 Optimization Certificate

For completeness, fix a decoder g and write $b_{y,T,a,s}$ for the support-function epigraph with score $Tv_{M,a,s}$, $c_{a,s} \geq v_{M,a,s}(c)$ for every c, and $d_{s,y,T}$ for the utility support epigraph with score $TM\ell_y$. Each b and d is represented by the dual in Eq. (6); its score is linear in M. The transform-exact LP is

$$\begin{aligned}
&\min_{M,\zeta} \quad \zeta && (34)\\
&\text{s.t.} \quad M_{cz} \geq 0, \quad \sum_z M_{cz} = 1 && \forall c,\\
&\qquad \frac{1}{G}\sum_s [(1-\eta_y)b_{y,T,a,s} + \eta_y c_{a,s}] \leq \tau_{P,y}^{\mathrm{BA}} && \forall y, T, a,\\
&\qquad (1-\eta_y)d_{s,y,T} + \eta_y u_y^{\max} \leq \zeta && \forall s, y, T,
\end{aligned}$$

where $u_y^{\max} \geq (M\ell_y)(c)$ for every c and $\tau_{P,y}^{\mathrm{BA}} = G^{-1} + (1 - G^{-1})\tau_{P,y}$. All epigraph variables are minimized indirectly or upper-constrained, so an optimum can take each at its smallest valid value. Enforcing every finite transform extreme and all G^L attacker assignments exactly implements the maxima in the certificates. Enumerating all J^L decoders and taking the smallest objective returns the global transform-exact finite-alphabet solution.

For the capacity fallback, each expression $\min\{\mathrm{Cap}_G(M), \overline{A}_y + \rho_y\}$ and $\min\{U_y, \overline{E}_{s,y} + \kappa_y\}$ is represented by a binary disjunction. With branch bit b, the first branch constraint is relaxed by $2b$ and the second by $2(1-b)$; all constituent quantities lie in ranges for which two is a valid big-M constant. The implementation analogously disjoins the utility branches, then enumerates decoders. Thus the fallback is a finite MILP, not a local nonlinear search.

Globality is asserted only when every decoder and attacker assignment is enumerated, the exact LP reports an optimal status or the fallback MIP gap is at most 10^{-10}, and the solver feasibility tolerance is 10^{-9}. The original bridge run preserves raw solver outputs, but claim-grade decisions come from a pre-inspection strict replay triggered by an adversarial proof audit. That replay contracts retained mass using a 10^{-9} outward guard until every independently recomputed membership slack is nonnegative, optimizes source constraints with a 10^{-6} inward margin, adds 10^{-9} to recomputed risk bounds, and compares those outward values with the original thresholds using zero decision tolerance. A timeout, negative membership slack, oversized assignment family, or post-solve mismatch can only abstain.

The exact audit checks zero-radius agreement with a separate population evaluator, binary confidence-ball endpoints, 100 randomized dominance comparisons with the fallback, and dense channel grids. The fallback audit compares 30 randomized small instances with exhaustive 21^K grids and includes a witness in which the best stochastic certified error is 0.371588 while every deterministic release has certified error 1.

12 Locked Experimental Protocol

The synthetic population has two labels, two sources, and three fine tokens. For label zero the source laws are $(0.80, 0.15, 0.05)$ and $(0.65, 0.30, 0.05)$; for label one they are $(0.05, 0.20, 0.75)$ and $(0.05, 0.35, 0.60)$. The common-transform set is the convex hull of identity and a nearest-neighbor smoothing channel. The hard safety scenario uses $\eta = 0.20$, source-advantage threshold 0.30, and worst conditional error threshold 0.40. The retention scenario uses $\eta = 0.10$, thresholds 0.35 and 0.45. Every source–label stratum contains exactly n independent draws.

The confirmation fixes 1,000 replicates in each of three hard-scenario and five retention-scenario sample-size cells. It compares eight rules and stores 64,000 method-level receipts. The preregistration fixes all seeds, methods, intervals, and pass criteria, hashes the theorem and implementation, and was pushed before the confirmation began. Pilot seeds zero through 199 are excluded. The oldest aggregate-only pilot is retained as design history but cannot support a channel-level claim; every later pilot row is replayed by the exact evaluator.

Matched finite-family baselines. A separately locked extension evaluates the same 1,000 paired tables per scenario with a 5^3 grid of binary-output stochastic channels and all 2^2 binary decoders, for 500 channel–decoder pairs. The confidence-region grid applies MOSAIC's one simultaneous pre-channel event but restricts optimization to these 500 pairs. The deterministic baseline applies the same shift-aware certificate to every deterministic three-to-two map. This is a matched test of FARE's restricted finite-encoder principle, not an implementation of FARE's raw-input fair-tree training pipeline or demographic-parity objective.

The LTT baseline tests every fixed pair without reusing MOSAIC's table region. Index its source-attacker and source–label utility inequalities by j. After removing the known contamination term, each has an empirical mean $\widehat{\mu}_j$, unsafe null $H_j : \mu_j \geq b_j$, and effective sample size n_j (or Gn for a balanced attacker average over G source strata). Its one-sided Hoeffding value is

$$p_j = \exp\{-2n_j(b_j - \widehat{\mu}_j)_+^2\}. \tag{35}$$

The unsafe candidate null is the union of its component nulls, so $p_{\mathrm{cand}} = \max_j p_j$ is valid by an intersection–union test: whenever any H_j holds, rejection at level α implies rejection of that true component at level α. Holm step-down then controls familywise false certification across all 500 candidates at $\delta = .05$. Among rejected candidates, LTT selects the smallest point-estimated worst error with a fixed index tie break; no rejection means abstention. The population evaluator grades all four rules on the paired untouched law, including every abstention and false acceptance.

The data-certified bridge confirmation was locked separately after excluding all real-study seeds through 1199. It runs seeds 1200–1219 on Waterbirds, Camelyon17-WILDS, CivilComments-WILDS, BiasBios-Clinical, and GaitPDB. Within each represented target source–label stratum, a deterministic seeded split assigns two thirds to the bridge and one third to an untouched diagnostic set. The frontier has identity plus twelve settings from pinned official INLP, LEACE, R-LACE, TaCo, and MANCE++ repositories, with no proxy implementations. Each of 13 candidates has one reference and one bridge table; familywise $\delta = 0.05$ is divided across these 26 tables, and each table's allocation is divided across its four source–label strata. Thus each multinomial radius uses $\delta/(13 \cdot 2 \cdot 4)$. The same event covers all bridge transforms, two- and four-token release optimizers, and utility thresholds $\{0.30, 0.35, 0.40, 0.45, 0.49\}$.

The primary rule uses $L = 2$, maximum source advantage 0.35, and maximum worst-stratum error 0.40. It selects the feasible candidate with minimum bridge-certified error and a lexical tie break. The minimum-error candidate is then re-optimized at $L = 4$ without inspecting diagnostics. All 100 dataset–seed jobs, 1,300 candidate rows, and 1,400 global optima are required; the stopping rule forbids outcome-based replacement or omission. Confirmatory gates require zero optimization errors, zero primary diagnostic contract violations, at least 16 estimable primary deployments, BiasBios deployment on at least 16/20 seeds at threshold 0.40 and 14/20 at 0.35, pointwise noninferiority of the globally optimized $L = 4$ objective, and zero retained mass for every Camelyon job with missing bridge support. The claim is restricted to populations drawn i.i.d. from the same target stratum laws as the bridge fold and is not a clinical-safety claim.

After 27 of 100 raw receipts had been written, but before any receipt contents or outcomes were inspected, an adversarial proof audit identified that the original tolerance-based acceptance rule did not justify an exact software contract. A hash-locked numerical amendment therefore supersedes every raw bridge decision with the strict replay above. The amendment records the timing, the information seen (filenames, counts, timestamps, and process status only), all code hashes, the zero-tolerance decision rule, and the requirement to replay all 1,400 optima regardless of outcome. Both raw and strict receipts remain public so this correction cannot be hidden.

13 Confirmation Receipts and Complete Primary Cells

The locked preregistration has SHA-256 digest `d6819cec1605db6dfeb5789c0f0f8d67` `4ee135b51f1cd820aaff16df84b65cf2`. The 45 MB confirmation receipt has digest `de4fd53da17e0371d9023ef5a58f2cf3` `093ed3111b980920e0f9b9eab1b4e665`. Its separate replay checks all 64,000 seeds and channels, 192,000 stochastic rows, 128,000 exact risks, 128,000 decision labels, and 1,152 aggregate fields; the maximum replay discrepancy is exactly zero in double precision. The replay itself has digest `9e64b4883cfb5bf510a058e0fe6995abb` `e362b2d958c0d1163fbaca78d2966d4`.

For zero events in 1,000 trials, the exact 95% Clopper–Pearson upper endpoint is 0.3682%. MOSAIC's retention rate has interval $[53.16, 59.40]$; held-out certification has $[1.62, 3.67]$ and finite LTT has $[2.28, 4.60]$. The theory alignment audit links both pre-outcome and post-outcome receipts and passes with mean absolute deployment error 0.001968 and primary-cell error 0.004389. The finite-sample concentration abstention bound is vacuous through $n = 500$, then falls to 0.03097 at $n = 1000$ and 7.60×10^{-9} at $n = 2000$; we report that looseness rather than replacing it with the local approximation.

14 Transform-Exact Confirmation and Audit Amendment

The strict-decision v2 preregistration has SHA-256 digest `89243fea30a158213679ed310e232e97c` `874137a57d97cccd4f6f07c6ce4b867`. Its full 10,000-row report has digest `80c5c77a5d98eba16a4a9a5d60f38871` `8115ef600546bc766aec106eebfb1421`, and the independent audit digest is `b75a99e90b8e28e042cf2b1e5ec0ce72` `651dc44f0b015b0b7c6cacda71ce78be`.

Declared numerical amendment. The original v1 confirmation passed every pre-audit gate, but its independent

Rule	Hard safety boundary, $n = 125$ Deploy	False	Safe	Retention regime, $n = 250$ Deploy	False	Safe
Always deploy plug-in	100.0	57.6	42.4	100.0	0.0	100.0
Plug-in continuum	73.7	42.1	31.6	100.0	0.0	100.0
Shift-unaware MOSAIC	80.3	76.6	3.7	100.0	0.0	100.0
Held-out fixed	0.0	0.0	0.0	2.5	0.0	2.5
Finite LTT	0.0	0.0	0.0	3.3	0.0	3.3
Deterministic MOSAIC	0.0	0.0	0.0	0.0	0.0	0.0
MOSAIC	0.0	0.0	0.0	56.3	0.0	56.3
Population oracle	100.0	0.0	100.0	100.0	0.0	100.0

Table 1: All eight locked rules in both primary cells (rates in percent). The oracle verifies that a safe channel exists; it is not deployable.

replay marked two of 10,000 deployment decisions as mismatches. All 80,000 recomputed certificate and external-risk quantities matched. The two returned capacity-fallback channels exceeded the source-leakage threshold by 4.12×10^{-10} and 4.40×10^{-10}: below the solver's 10^{-9} feasibility and 2×10^{-7} post-hoc tolerances, but above the replay's 10^{-10} decision tolerance. We preserved the failed audit, changed the deployment layer to recheck every source-leakage certificate at 10^{-10}, and recorded before rerunning that exactly those two rows should flip to abstention without changing any primary gate. The hash-locked v2 rerun matched that prediction exactly; its audit has zero mismatch.

The confidence event holds on 999 of 1,000 tables in the hard cell, on all retention $n = 125$ tables, on 999 tables at $n = 250$ and 500, and on all tables at $n = 1000$. There are no failures on the confidence event and no observed false acceptances off it. For zero events in 1,000 trials, the exact 95% binomial upper endpoint is 0.3682% per method–cell; this empirical endpoint is distinct from the theorem's familywise $\delta = 0.05$ guarantee.

The exploratory five-threshold receipt has digest `5147a0a73ce888265351baa8ee638b75` `1fc4837b60f96335d1fccd14442bdeea`. It reuses the audited channels and exact population risks, holds the source-leakage threshold at .35, and changes only the utility decision threshold. Because four thresholds were selected after outcomes, the sensitivity table in the main paper is explicitly descriptive. All 20 method–size–threshold cells have zero false acceptance, and exact retention is never lower than capacity transfer.

15 Transform-Exact Verification Matrix

The exact support implementation is checked in three independent ways. At zero radius, source leakage, and utility match a population evaluator that enumerates every transform extreme, attacker assignment, and residual simplex vertex. For binary fine alphabets, confidence-ball endpoint enumeration reproduces the support-function result. Finally, 100 randomized instances verify that the exact envelope never exceeds the capacity fallback. The optimizer is then checked against dense channel grids and an independent post-solve certificate recomputation. Eleven focused transform-exact tests pass as part of a repository-wide run of 158 tests plus 14 subtests.

16 Official-Method Real-Feature Receipts

Fresh paired confirmation. The untouched seeds are 105–109; pilots 0–4 and the earlier real-feature seeds 100–104 are excluded. The protocol was locked before these outcomes with SHA-256 digest `d88d42e813d65c7d725e923c94be0518` `d7049e0b62131be96ffbec748d5e5775`. All 25 jobs and 325 candidate pairs complete with official INLP, LEACE, R-LACE, TaCo, and MANCE++ implementations and no proxy or failed row. All ten registered gates pass.

The frontier order is identity; INLP ranks 1, 2, 4, and 8; LEACE; R-LACE ranks 1 and 4; TaCo with 1, 2, 3, and 5 removed components; and MANCE++. Every candidate stores its pinned repository, commit, entry point, and parameters. Each separately fitted tokenizer receives $\delta/13$; both certificate variants optimize a continuum of channels from the same covered table.

Across the 325 paired objectives, transform-exact is no worse on every row and strictly better on 135, with mean, median, and maximum gains of .01036, 2.6×10^{-15}, and .03451. The effectively zero median reflects exact ties on the 190 rows for which the fallback is already tight. Candidate deployments increase from 130 to 131. The only decision-status crossing is Waterbirds seed 107, which is externally estimable and safe. All five BiasBios selections remain safe. CivilComments and GaitPDB each abstain on five seeds. Camelyon17 contributes five model-conditional deployments but no estimable external diagnostic, exactly matching the missing-source theorem.

The independent audit recomputes every confidence radius, both certificate families, both selected frontiers, all external diagnostics, and all 650 global optimization solutions. It reports zero mismatches, pointwise dominance, and no failures. The manifest digest is `c12f18c9b52bb68522691386131faad9` `66c1ae60d9cb0c540af8b909dd1eecc8`; the audit digest is `becebaa2839c776fcd806d3d8222a1e2` `8c213259300a7781a124bf3feb8fd600`.

Preserved predecessor evidence. An earlier lock on seeds 100–104 evaluated only the capacity fallback: five BiasBios selections were estimable and safe, five Camelyon17 selections were unestimable, and 15 jobs abstained. After those outcomes were known, an explicitly exploratory exact replay improved 133 of 325 rows and added two safe Waterbirds

Scenario	n	Safe deploy (%)		Mean cert. error		Mean gain (pp)	False
		transfer	exact	transfer	exact		accepts
Hard boundary	125	0.0	0.0	.5004	.4959	0.45	0/2000
Retention	125	0.3	53.0	.4808	.4482	3.26	0/2000
Retention	250	57.8	99.9	.4479	.4070	4.09	0/2000
Retention	500	100	100	.4162	.3799	3.63	0/2000
Retention	1000	99.8	100	.3888	.3612	2.77	0/2000

Table 2: Every locked transform-exact confirmation cell. Each method has 1,000 paired tables per row. The exact objective is no larger on all 5,000 pairs and strictly smaller on all 4,000 retention pairs.

Dataset/seed	Exact selection	Cert.	Ext. error	Ext. adv.
BiasBios 105	R-LACE rank 1	.2969	.1902	.0579
BiasBios 106	INLP rank 4	.2882	.2143	.0380
BiasBios 107	INLP rank 4	.2920	.2011	.0711
BiasBios 108	R-LACE rank 4	.2952	.2390	.0610
BiasBios 109	INLP rank 4	.2814	.1881	.0082
Waterbirds 107	INLP rank 4	.4651	.2905	.0569

Table 3: Every externally estimable transform-exact selection. All six pass the natural external diagnostic.

Seed	Fallback selection	Exact selection	Exact cert.
105	LEACE	INLP rank 1	.2598
106	LEACE	LEACE	.2302
107	LEACE	LEACE	.2492
108	LEACE	LEACE	.2590
109	INLP rank 1	INLP rank 1	.2432

Table 4: Camelyon17 selections. Every natural external diagnostic is unestimable because a required source class is absent.

deployments. That result motivated the fresh paired protocol above; it is not pooled with or substituted for the new confirmation. The earlier preregistration, manifest, fallback audit, exploratory analysis, and exploratory audit remain in the anonymous artifact for a complete record.

As throughout the paper, external safety here is descriptive. The zero false acceptances among six estimable selections has a two-sided 95% Clopper–Pearson upper bound of 45.9%, and a favorable benchmark split cannot establish membership in the structured shift class.

17 Reproducibility Manifest

MOSAIC optimization, confirmations, and audits ran on an Apple M4 system with 10 CPU cores and 16 GB unified memory under macOS 26.2 (arm64). The locked Python environment was Python 3.12.13 with NumPy 2.5.1, SciPy 1.18.0, scikit-learn 1.9.0, PyTorch 2.13.0, and the HiGHS LP/MILP backend exposed by SciPy. No GPU is required once the registered feature stores have been constructed; synthetic confirmation and every replay are CPU-only.

The named package reports the exact preregistration hash, Git commit, software versions, and artifact hashes. The anonymous package retains the same internal identifiers while replacing public authorship and repository metadata as required by double-blind review. Raw synthetic receipts contain no private or human-subject data. Real-domain experiments, if included, retain dataset terms, official split definitions, preprocessing hashes, and method-specific code provenance in a separate manifest.